

Task Agnostic and Task Specific Self-Supervised Learning from Speech with LeBenchmark

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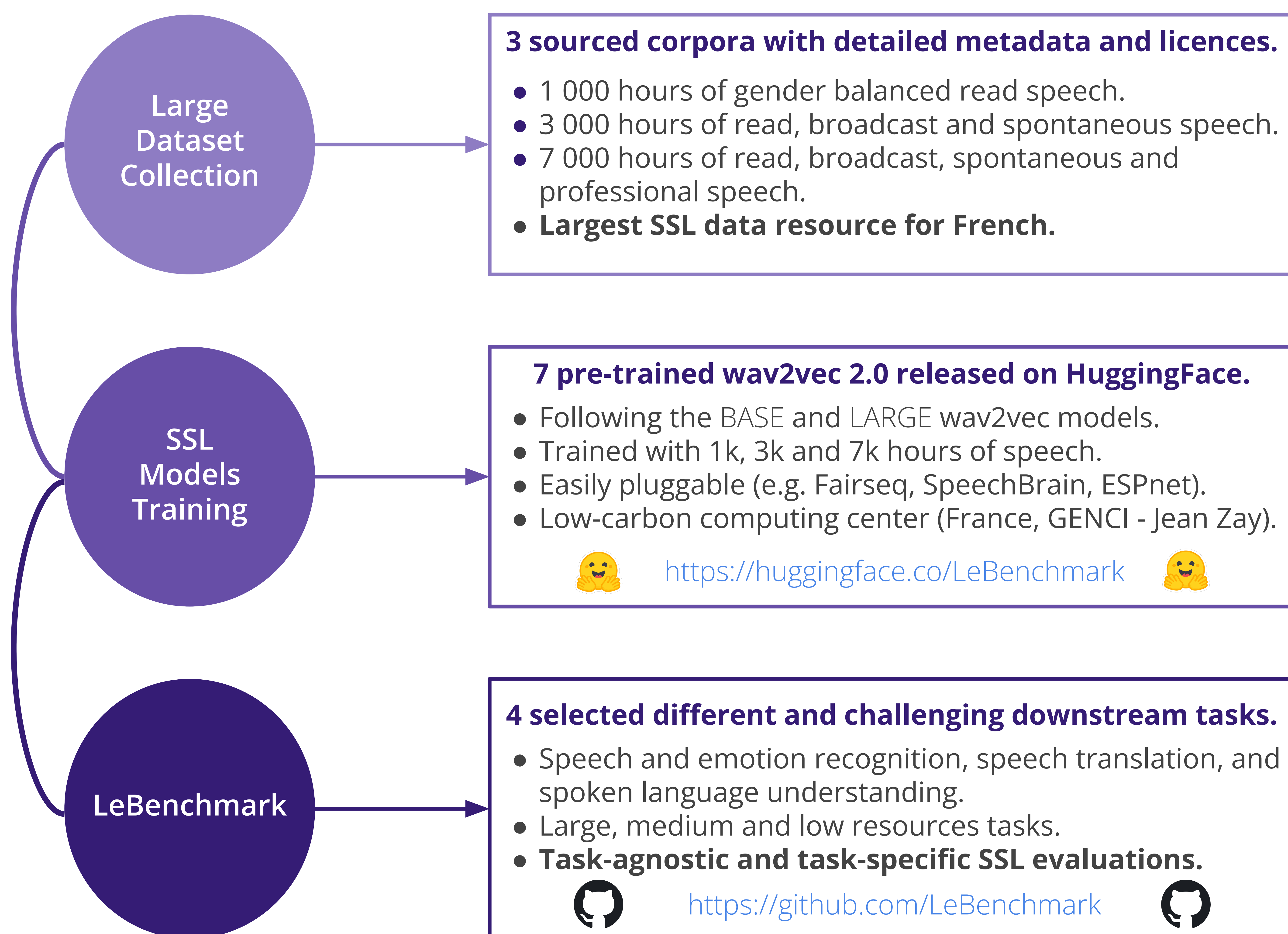
Motivations

- Self supervised learning (**SSL**) models on **speech** are promising.
- **Lack of standardization** in the evaluation process for comprehensive comparisons of these models.
- Issue gets even worse with the investigation of SSL approaches for **other languages than English**.

➡ We introduce and provide a **reproducible benchmark!**



Contributions



Takeaways

- SSL improves over conventional acoustic features for **all tasks**.
- **Massive improvements** for **medium-resources** downstream tasks.
- **Task-specific pre-training** further **improves the performance**.
- For non-ASR tasks (SLU, AST), **SSL fine-tuning for ASR on in-domain data is beneficial**.